MONASH University

MONASH BUSINESS SCHOOL

ETF3231/5231 Business forecasting

Ch7. Regression models

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Monash University CRICOS Provider Number: 00008C

Outline

1 The linear model with time series

2 Some useful predictors for linear models

3 Residual diagnostics

4 Selecting predictors and forecast evaluation

5 Forecasting with regression

6 Matrix formulation

7 Correlation, causation and forecasting

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Multiple regression and forecasting

$$y_t = \beta_0 + \beta_1 x_{1,t} + \beta_2 x_{2,t} + \cdots + \beta_k x_{k,t} + \varepsilon_t.$$

- y_t is the variable we want to predict: the "response' ' variable
- Each $x_{j,t}$ is numerical and is called a "predictor' '. They are usually assumed to be known for all past and future times.
- The coefficients $\beta_1, \ldots, \beta_k$ measure the effect of each predictor after taking account of the effect of all other predictors in the model.

That is, the coefficients measure the marginal effects.

- ε_t is a white noise error term

Example: US consumption expenditure

```r
fit_cons <- us_change %>%
  model(lm = TSLM(Consumption ~ Income))
report(fit_cons)
```

```
## Series: Consumption
## Model: TSLM
##
## Residuals:
##     Min      1Q Median      3Q     Max
## -2.582 -0.278  0.019  0.323  1.422
##
## Coefficients:
##             Estimate Std. Error t value Pr(>|t|)
## (Intercept)   0.5445     0.0540   10.08  < 2e-16 ***
## Income        0.2718     0.0467    5.82  2.4e-08 ***
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.591 on 196 degrees of freedom
## Multiple R-squared: 0.147,   Adjusted R-squared: 0.143
## F-statistic: 33.8 on 1 and 196 DF, p-value: 2e-08
```

Example: US consumption expenditure

```r
fit_consMR <- us_change %>%
  model(lm = TSLM(Consumption ~ Income + Production + Savings + Unemployment))
report(fit_consMR)
```

```
## Series: Consumption
## Model: TSLM
##
## Residuals:
##     Min      1Q Median      3Q     Max
## -0.906 -0.158 -0.036  0.136  1.155
##
## Coefficients:
##               Estimate Std. Error t value Pr(>|t|)
## (Intercept)    0.25311    0.03447    7.34  5.7e-12 ***
## Income         0.74058    0.04012   18.46  < 2e-16 ***
## Production     0.04717    0.02314    2.04    0.043 *
## Savings       -0.05289    0.00292  -18.09  < 2e-16 ***
## Unemployment -0.17469    0.09551   -1.83    0.069 .
## ---
## Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1
##
## Residual standard error: 0.31 on 193 degrees of freedom
## Multiple R-squared: 0.768,   Adjusted R-squared: 0.763
## F-statistic:  160 on 4 and 193 DF, p-value: <2e-16
```

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Trend

Linear trend

$$x_t = t$$

- $t = 1, 2, \ldots, T$
- Strong assumption that trend will continue.

Nonlinear trend

Piecewise linear trend with bend "knot" at τ

$$x_{1,t} = t$$

$$x_{2,t} = (t - \tau)_+ = \begin{cases} 0 & t < \tau \\ (t - \tau) & t \geq \tau \end{cases}$$

- β_1 trend slope before time τ
- $\beta_1 + \beta_2$ trend slope after time τ
- More knots can be added forming more $(t - \tau)_+$

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Quadratic or higher order trend

$$x_{1,t} = t, \quad x_{2,t} = t^2, \quad \ldots$$

NOT RECOMMENDED!

Uses of dummy variables

Seasonal dummies

- For quarterly data: use 3 dummies
- For monthly data: use 11 dummies
- For daily data: use 6 dummies
- What to do with weekly data?

Outliers

- If there is an outlier, you can use a dummy variable to remove its effect.

Holidays

For monthly data

- Christmas: always in December so part of monthly seasonal effect
- Easter: use a dummy variable $v_t = 1$ if any part of Easter is in that month, $v_t = 0$ otherwise.
- Ramadan and Chinese new year similar.

For daily data

- If it is a public holiday, dummy=1, otherwise dummy=0.

Fourier series

Periodic seasonality can be handled using pairs of Fourier terms:

$$s_k(t) = \sin\left(\frac{2\pi kt}{m}\right) \qquad c_k(t) = \cos\left(\frac{2\pi kt}{m}\right)$$

$$y_t = a + bt + \sum_{k=1}^{K} [\alpha_k s_k(t) + \beta_k c_k(t)] + \varepsilon_t$$

- Every periodic function can be approximated by sums of `sin()` and `cos()` terms for large enough K.
- Choose K by minimizing AICc.
- Called "harmonic regression"

```
TSLM(y ~ trend() + fourier(K))
```

Distributed lags

Lagged values of a predictor.

Example: x is advertising which has a delayed effect

$$
\begin{aligned}
x_1 &= \text{advertising for previous month;} \\
x_2 &= \text{advertising for two months previously;} \\
&\vdots \\
x_m &= \text{advertising for } m \text{ months previously.}
\end{aligned}
$$

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Multiple regression and forecasting

For forecasting purposes, we require the following assumptions:

- ε_t are uncorrelated and zero mean
- ε_t are uncorrelated with each $x_{j,t}$.

It is useful to also have $\varepsilon_t \sim \mathrm{N}(0, \sigma^2)$ when producing prediction intervals or doing statistical tests.

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Comparing regression models

- R^2 does not allow for “degrees of freedom’ ’.
- Adding *any* variable tends to increase the value of R^2, even if that variable is irrelevant.

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To overcome this problem, we can use *adjusted* R^2:

$$\bar{R}^2 = 1 - (1 - R^2)\frac{T - 1}{T - k - 1}$$

where k = no. predictors and T = no. observations.

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where k = no. predictors and T = no. observations.

Maximizing $\bar{R}^2$ is equivalent to minimizing $\hat{\sigma}^2$.

$$\hat{\sigma}^2 = \frac{1}{T-k-1}\sum_{t=1}^{T}\varepsilon_t^2$$

Akaike's Information Criterion

$$\text{AIC} = -2\log(L) + 2(k+2)$$

- L = likelihood
- k = # predictors in model.
- AIC penalizes terms more heavily than $\bar{R}^2$.

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$$\text{AIC}_\text{C} = \text{AIC} + \frac{2(k+2)(k+3)}{T-k-3}$$

- Minimizing the AIC or AICc is asymptotically equivalent to minimizing MSE via **leave-one-out cross-validation** (for any linear regression).

Bayesian Information Criterion

$$\text{BIC} = -2\log(L) + (k+2)\log(T)$$

where L is the likelihood and k is the number of predictors in the model.

- BIC penalizes terms more heavily than AIC
- Also called SBIC and SC.
- Minimizing BIC is asymptotically equivalent to leave-v-out cross-validation when $v = T[1 - 1/(log(T) - 1)]$.

Leave-one-out cross-validation

For regression, leave-one-out cross-validation is faster and more efficient than time-series cross-validation.

- Select one observation for test set, and use *remaining* observations in training set. Compute error on test observation.
- Repeat using each possible observation as the test set.
- Compute accuracy measure over all errors.

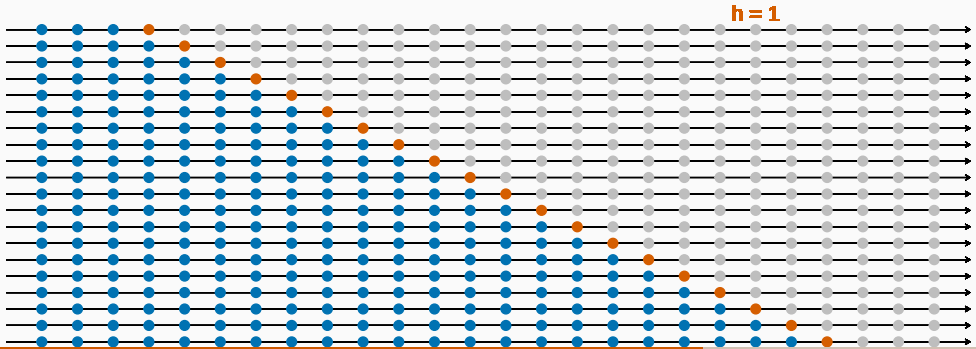
Cross-validation

Traditional evaluation

Cross-validation

Traditional evaluation

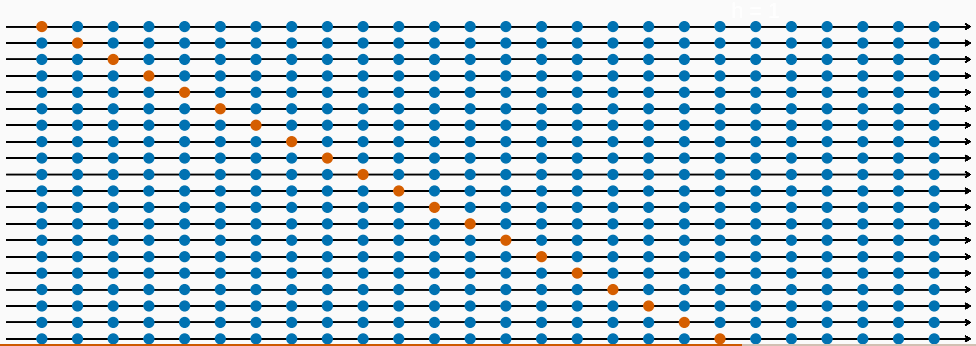
Time series cross-validation



Cross-validation

Traditional evaluation

Leave-one-out cross-validation



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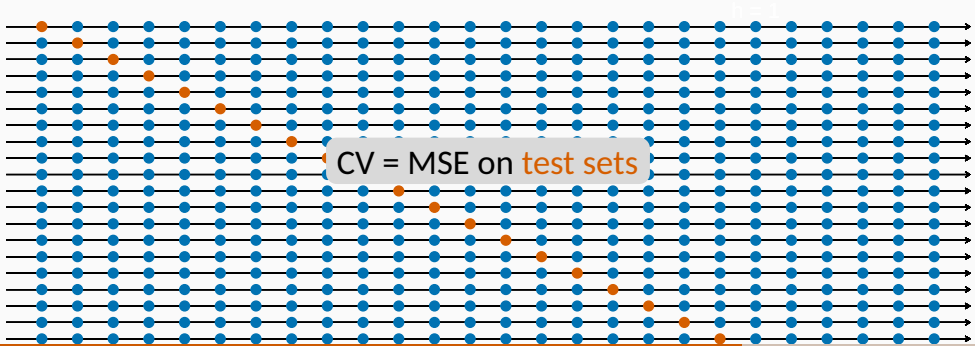

Choosing regression variables

Best subsets regression

- Fit all possible regression models using one or more of the predictors.
- Choose the best model based on one of the measures of predictive ability (CV, AIC, AICc).

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Warning!

- If there are a large number of predictors, this is not possible.
- For example, 44 predictors leads to 18 trillion possible models!

Choosing regression variables

Backwards stepwise regression

- Start with a model containing all variables.
- Try subtracting one variable at a time. Keep the model if it has lower CV or AICc.
- Iterate until no further improvement.

Forwards stepwise regression

- Start with a model containing only a constant.
- Add one variable at a time. Keep the model if it has lower CV or AICc.
- Iterate until no further improvement.

Hybrid backwards and forwards also possible.

- Stepwise regression is not guaranteed to lead to the best possible model.

What should you use?

Notes

- Stepwise regression is not guaranteed to lead to the best possible model.
- Inference on coefficients of final model will be wrong.

Choice: CV, AIC, AICc, BIC, $\bar{R}^2$

- BIC tends to choose models too small for prediction (however can be useful for large k).
- $\bar{R}^2$ tends to select models too large.
- AIC also slightly biased towards larger models (especially when T is small).
- Empirical studies in forecasting show AIC is better than BIC for forecast accuracy.

Choice between AICc and CV (double check AIC and BIC where possible).

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Ex-ante versus ex-post forecasts

- Ex ante forecasts are made using only information available in advance.
  - require forecasts of predictors
- Ex post forecasts are made using later information on the predictors.
  - useful for studying behaviour of forecasting models.
- trend, seasonal and calendar variables are all known in advance, so these don't need to be forecast.

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Multiple regression forecasts

Fitted values

$$\hat{\boldsymbol{y}} = \boldsymbol{X}\hat{\boldsymbol{\beta}} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'\boldsymbol{y} = \boldsymbol{H}\boldsymbol{y}$$

where $\boldsymbol{H} = \boldsymbol{X}(\boldsymbol{X}'\boldsymbol{X})^{-1}\boldsymbol{X}'$ is the "hat matrix' '.

Leave-one-out residuals

Let $h_1, \ldots, h_T$ be the diagonal values of $\boldsymbol{H}$, then the cross-validation statistic is

$$\text{CV} = \frac{1}{T}\sum_{t=1}^{T}[e_t/(1-h_t)]^2,$$

where e_t is the residual obtained from fitting the model to all T observations.

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Correlation is not causation

- When x is useful for predicting y, it is not necessarily causing y.
- e.g., predict number of drownings y using number of ice-creams sold x.
- Correlations are useful for forecasting, even when there is no causality.
- Better models usually involve causal relationships (e.g., temperature x and people z to predict drownings y).